

Experiment 4: Fixed Height + Obstacle + Multi Wind Strength | medium_wind | Comparison Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1981	2.323	1.308	4.888	35.7%	71.0%	82.8%	14.077	28.844	8.790	22.644	0.566	1.292	1.440
PPO-GRU	1992	3.094	0.772	13.414	63.0%	81.1%	82.8%	7.196	16.461	5.841	12.531	0.349	0.722	1.032
PPO-RNN	1990	1.966	1.074	2.913	46.8%	79.8%	90.5%	12.892	25.502	6.688	14.142	0.416	1.102	1.393
MPC	1060	5.901	3.094	15.604	3.6%	18.6%	46.8%	2.710	3.658	1.424	2.674	0.142	0.141	0.158

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1981	99.0%	19	0.9%	0.9%	19	100.0%	0	0
PPO-GRU	2000	1992	99.6%	8	0.4%	0.4%	8	100.0%	0	0
PPO-RNN	2000	1990	99.5%	10	0.5%	0.5%	10	100.0%	0	0
MPC	2000	1060	53.0%	940	47.0%	47.0%	940	100.0%	0	0

Cross-Case Check: PPO/PPO-GRU/PPO-RNN on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	940	933	99.3%	216	23.2%	2.672
PPO-GRU	940	934	99.4%	165	17.7%	3.309
PPO-RNN	940	932	99.1%	89	9.5%	2.006